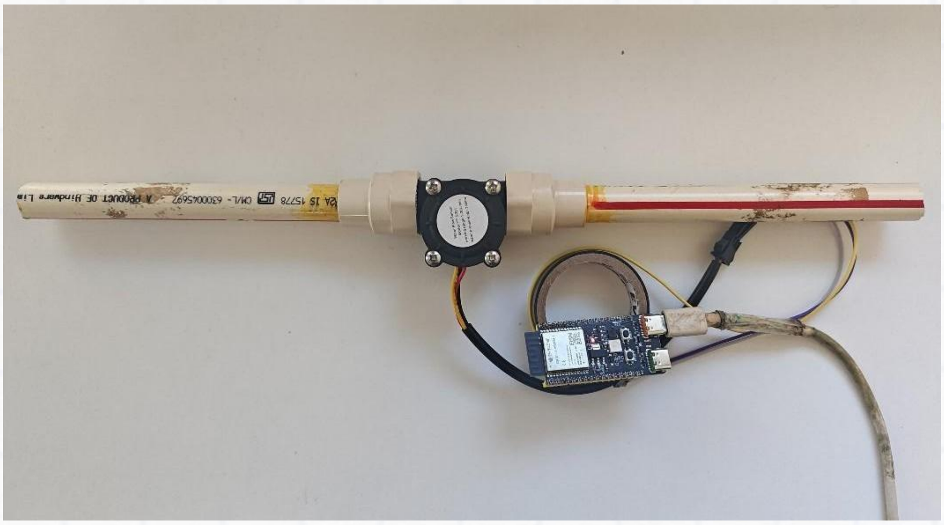
Smart Flow (Build a Flow Meter)

(An IoT-based water flow meter that measures and monitors real-time water usage parameters remotely)



Index

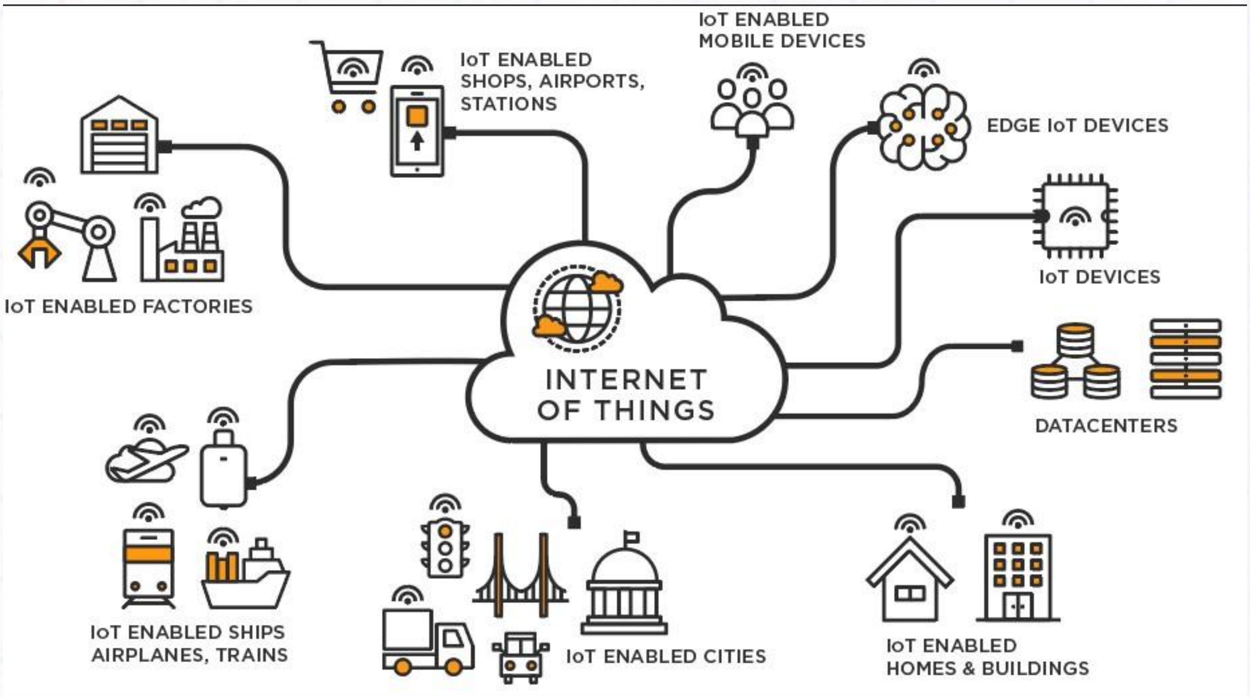
Outcome

Build an IoT (Internet of Things) Flow Meter that measures water flow rate and consumption in real-time and track this data remotely on a web IoT platform.

Concept

What is IoT?

IoT is a system of interconnected computing devices that use internet protocols to communicate and transfer data. This allows remote devices to communicate with each other without the need for human involvement



How this project works

In this project, a flow sensor measures the flow rate of any fluid passing through. This value is sent to the ESP32 board using an External interrupt mechanism - which constantly measures the flow of the fluid.
Through the code written to the ESP32, we calculate the values of the Flow rate and Water accumulated.

The ESP32 Wifi module then sends the data to a Blynk cloud server using an API. The water flow data can then be viewed in real-time on the Blynk web dashboard.

Components

Sno	Components	Quantity
1.	ESP32C6 Dev Module	1
2.	Water flow sensor (YF-S201)	1
3.	Jumper wires (Male to Female)	3
4.	USB to Type – C cable	1
5.	Funnel	1

Esp 32 C-6 Module

YF-S201 Flow sensor

Funnel

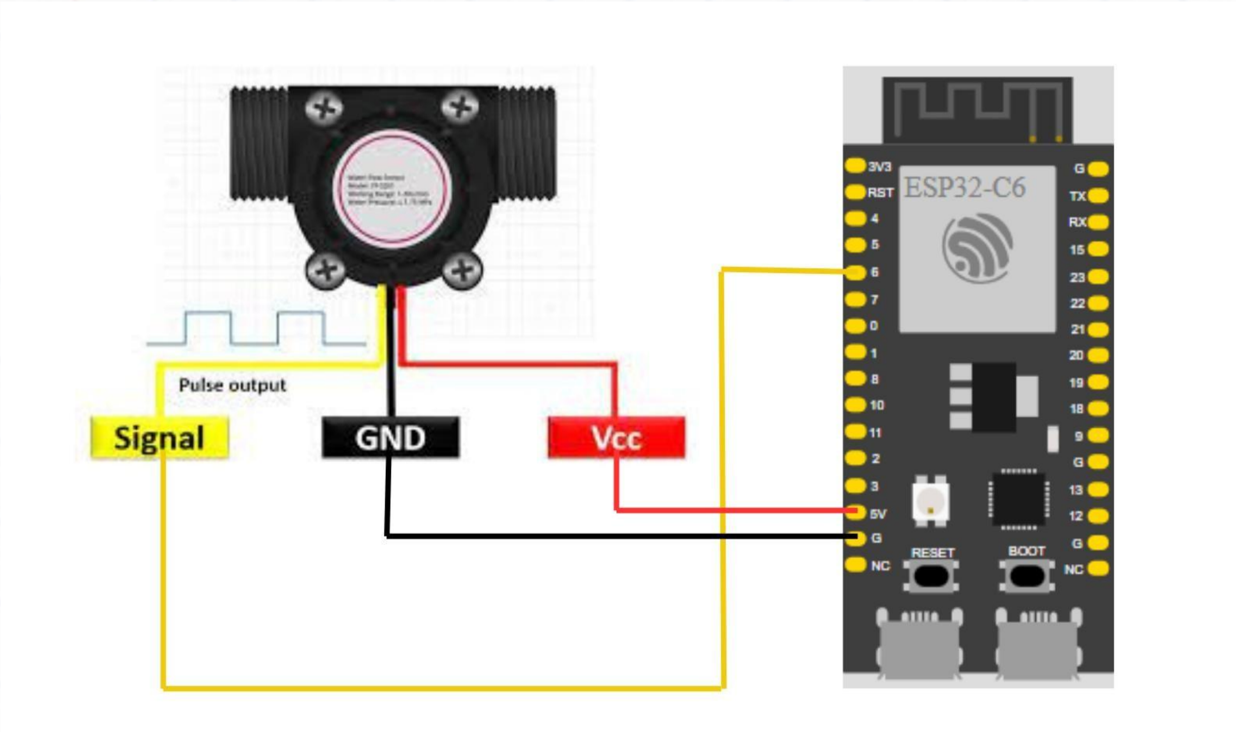
Jumper Wires

USB

Connections

Safety tip: Always ensure that the connections to the components are correct and completed before connecting the power supply to the Esp32.

Circuit Diagram



Follow the detailed steps in the following pages to complete the circuit.

NOTE: Before starting the connections, verify that all the jumper wires are working using a multimeter. Also ensure that the connections are strong, or else the setup may not work.

Detailed Connection Steps

Take 3 female-to-male jumper wires and connect them to the pins (5V, GND, D27) of the ESP 32 Board. Connect the other ends as per the below connections:
The YF-S201 Flow sensor has 3 wires RED, BLACK, and YELLOW

1. RED to 5v on ESP32
2. BLACK to GND on ESP32
3. YELLOW to D6 on ESP32

You are now ready to work on the software for the project.

Software

Launching the IDE for our project

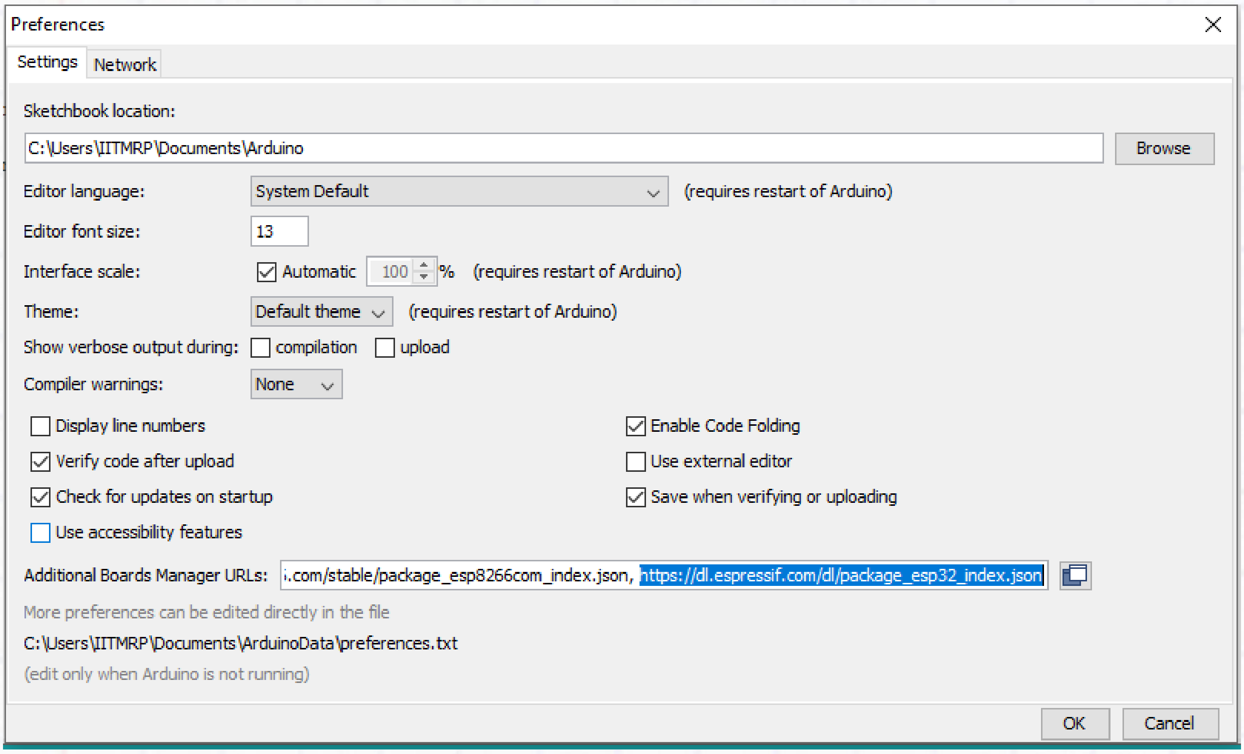
1. Install the Arduino IDE

If you haven't already installed the Arduino IDE, download and install it from the official Arduino website.

2. Install the ESP32 Board in Arduino IDE

1. Open the Arduino IDE.
2. Go to File > Preferences.
3. In the "Additional Board Manager URLs" field, add the following URL:

https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json



4. Go to Tools > Board > Boards Manager.
5. Search for ESP32 and install the ESP32 package. This will take some time to install.

3. Connect Your ESP32 Board

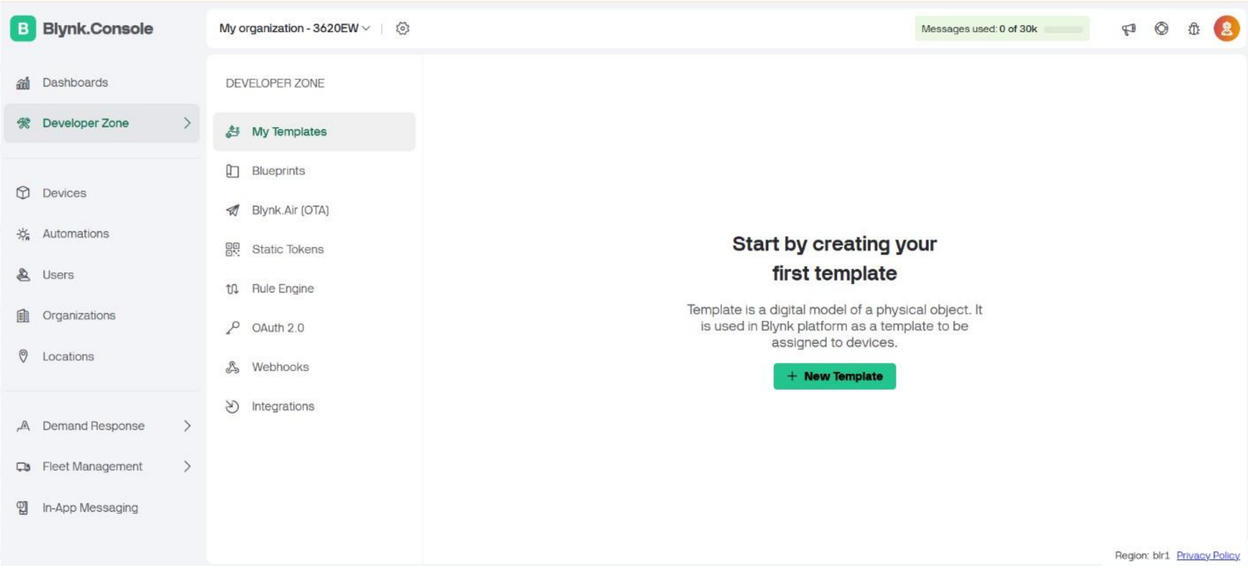
1. Connect your ESP32 board to your computer using a USB cable.
2. Go to Tools > Board > ESP32C6 Dev Module.
3. Go to Tools > Port and select the COM port to which your ESP32 is connected.

4. Prepare the Hardware

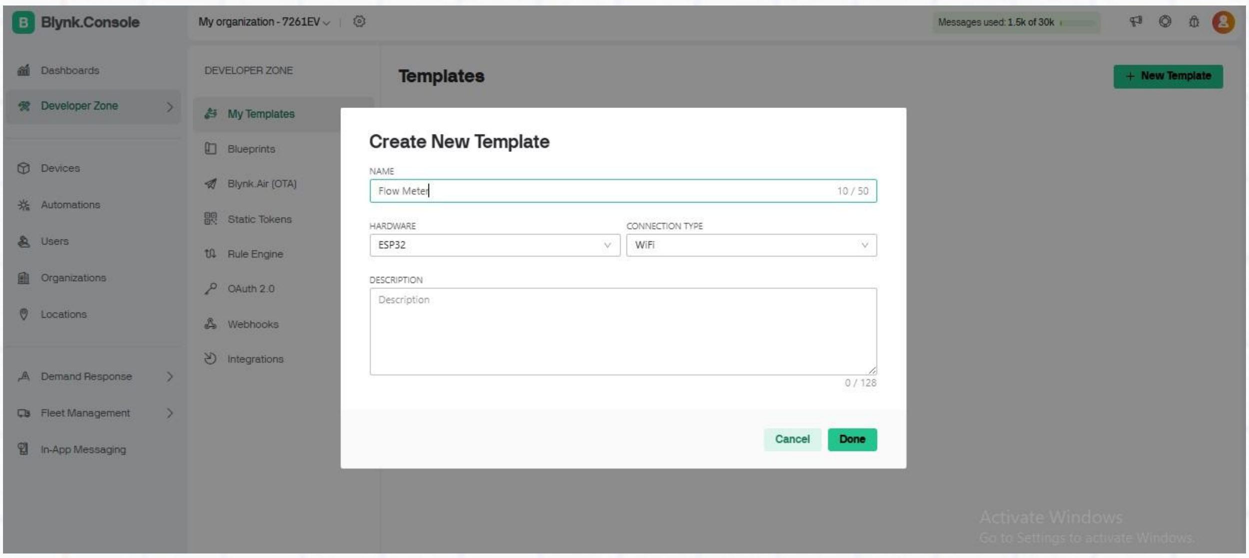
Connect your YF-S201 Flow sensor to the ESP32. All Three wires in the flow sensor are connected to the Esp32.

5. . Blynk Setup

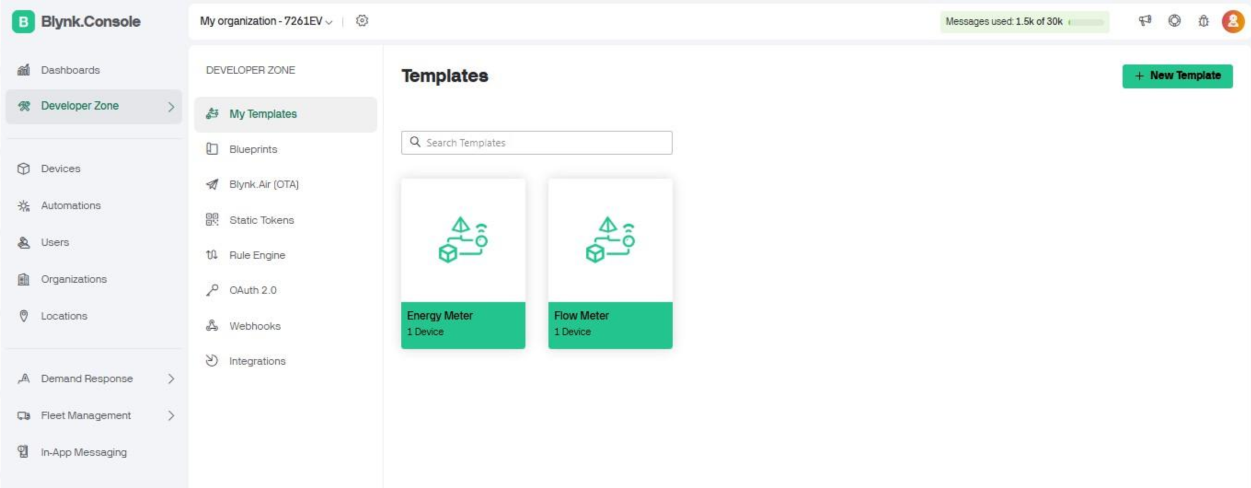
- For monitoring the water flow readings, we will be using the Blynk IoT Platform. Go https://blynk.io/ and sign up for free.
- On the blynk website, click on Developer Zone >> My templates as shown in the below.



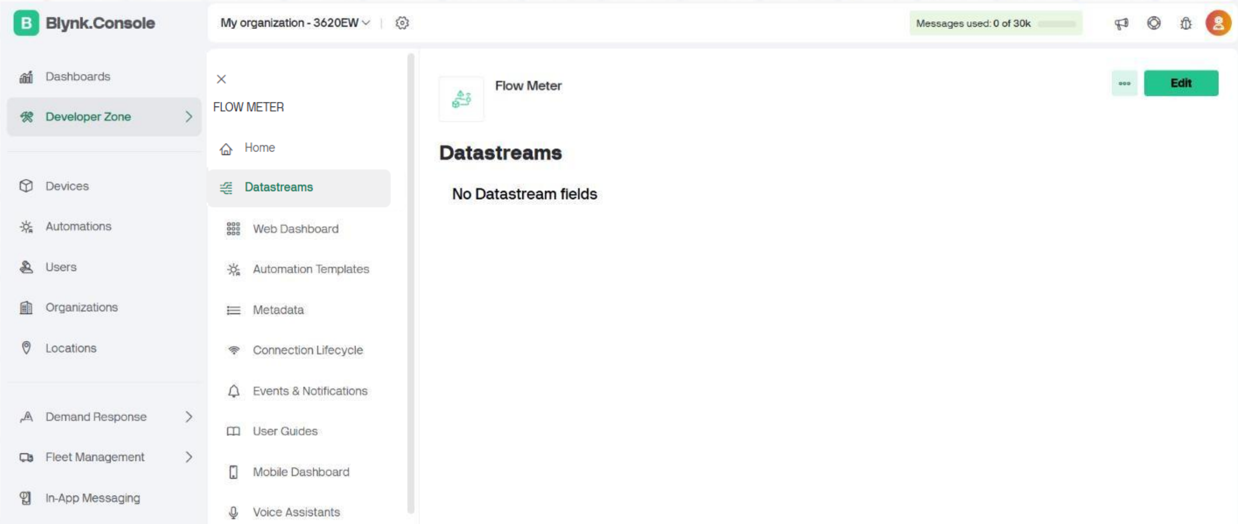
- Select **New Template** and give the Name of the template as **Flow meter** and check once the hardware is **ESP32** and the connection type is **WIFI** and click done.



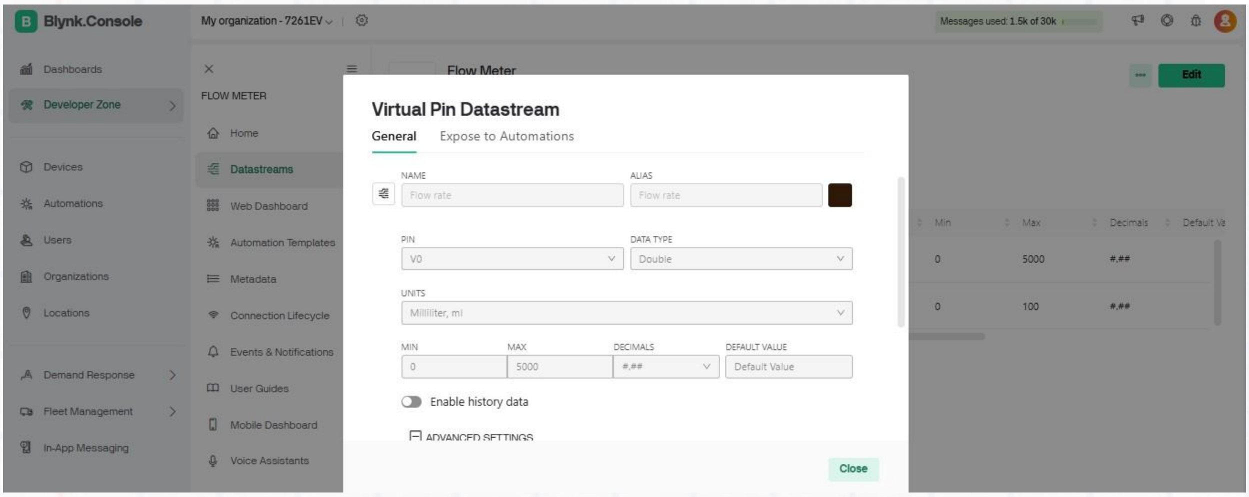
- Now click on Developer Zone >> My templates >> Flow Meter.



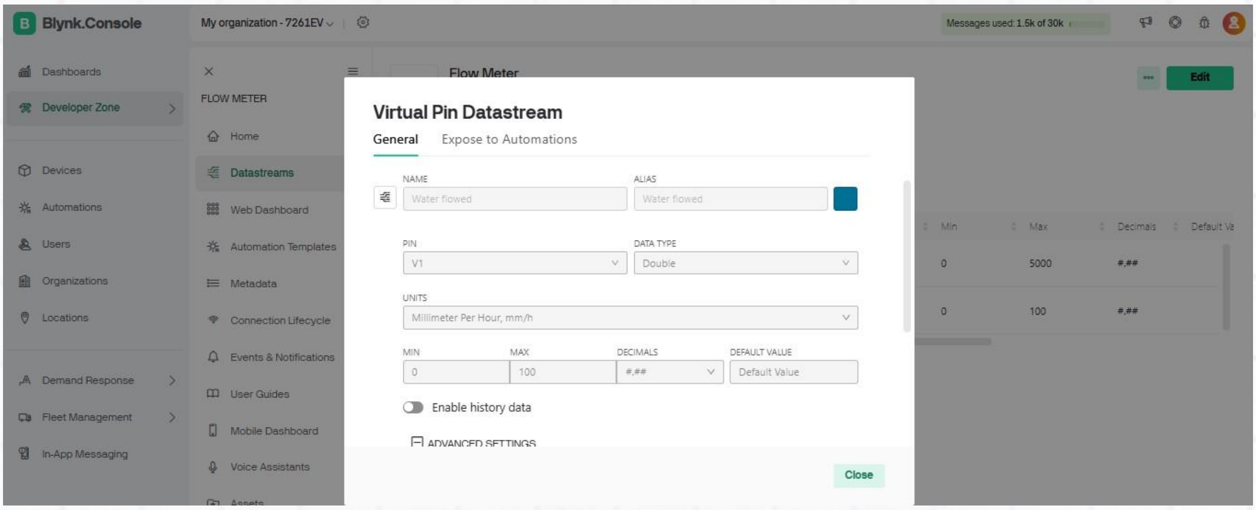
- Now click on Flow meter >> Data streams. After this click on Edit option on the top of the right corner.



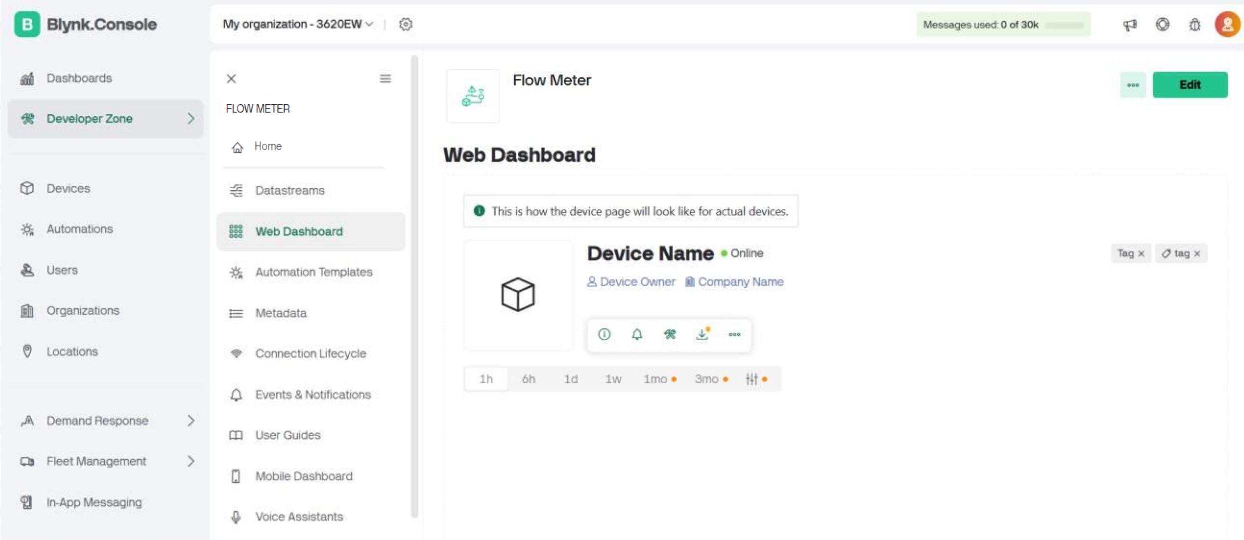
- Now click New DataStream >> virtual pin.
- Give parameters Name as Flow rate, Pin as v0, Data type as Double, Units as milli liters, Min as 0 and Max as 5000 and give then click create



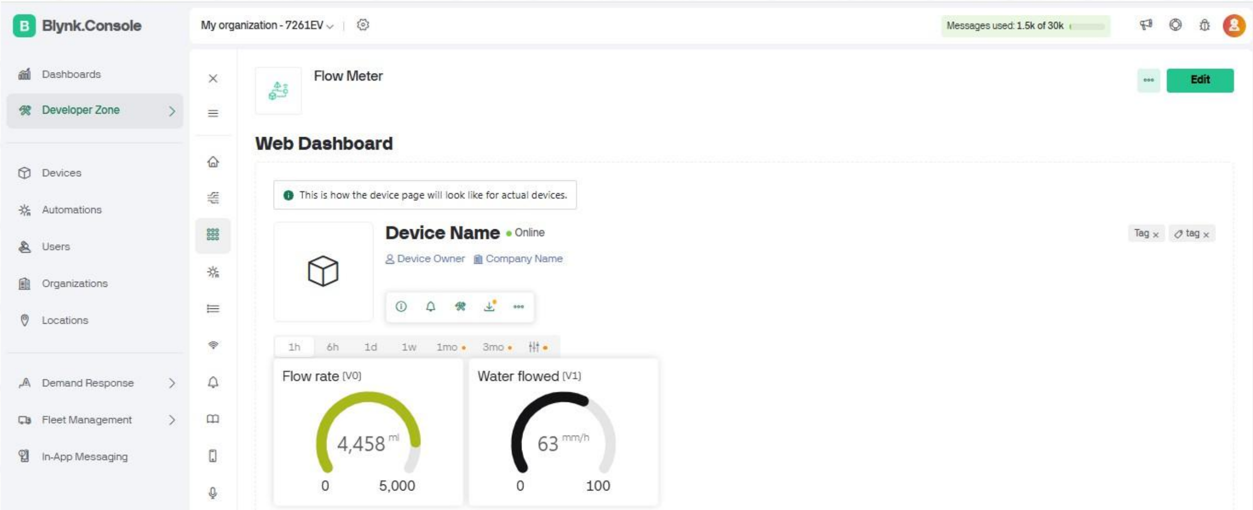
- Give parameters Name as Water flowed, Pin as v0, Data type as Double, Units as milli liter per hour, Min as 0 and Max as 100 and give then click create



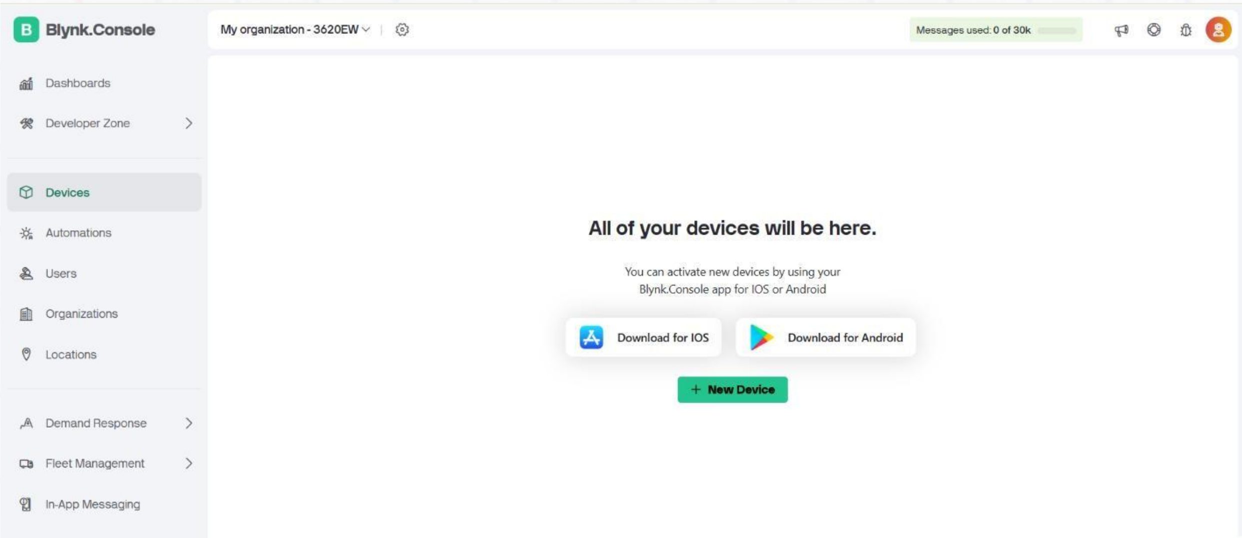
- After that click Developer zone >> My templates >> Flow Meter >> Web Dashboard on the left side.



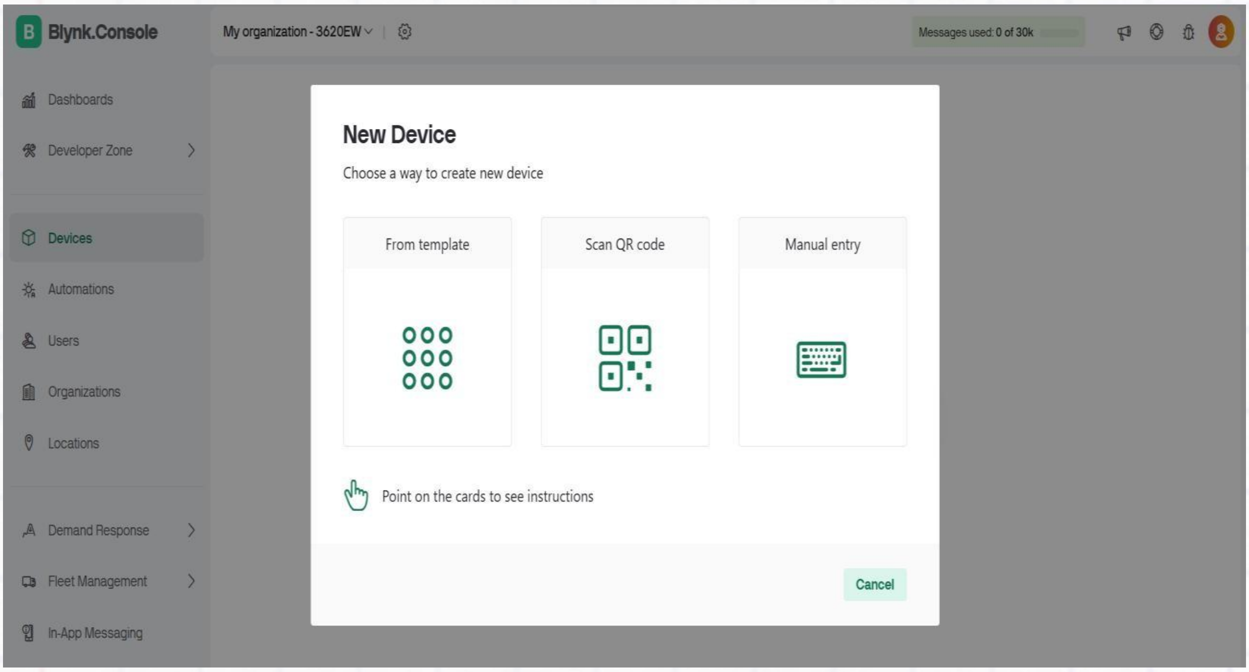
- Click Edit option in the top of the right side and scroll down the widget box double click Gauge.
- Click the gear icon ( ) in the gauge and give the title as Flow rate and data stream as Flow rate (V0), same process for Water flowed and save it.



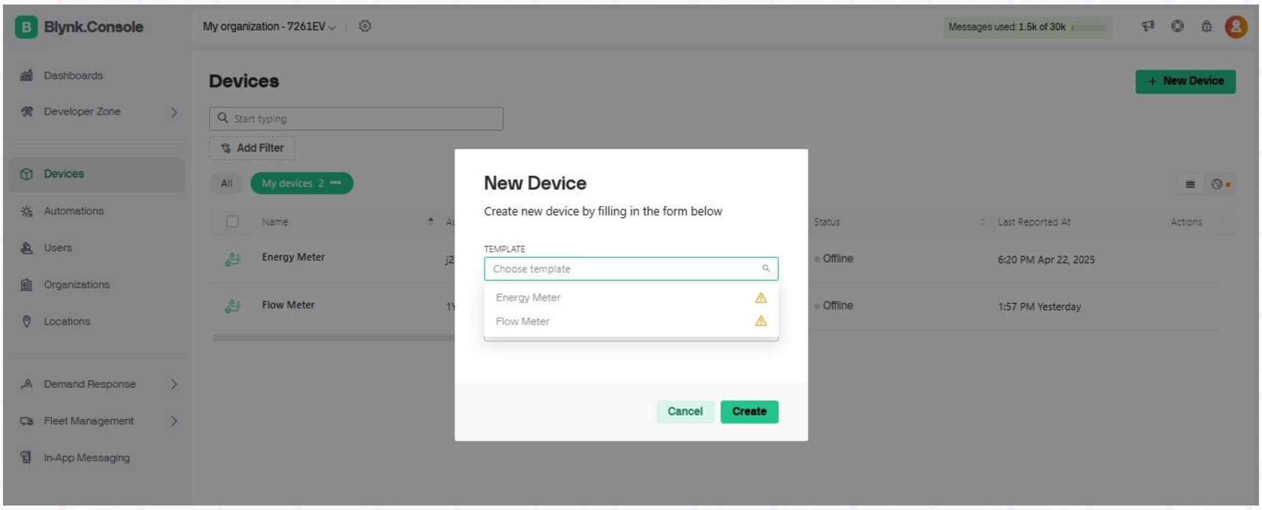
- Now Click Devices and Click New Device.



- Click New device and select From Template



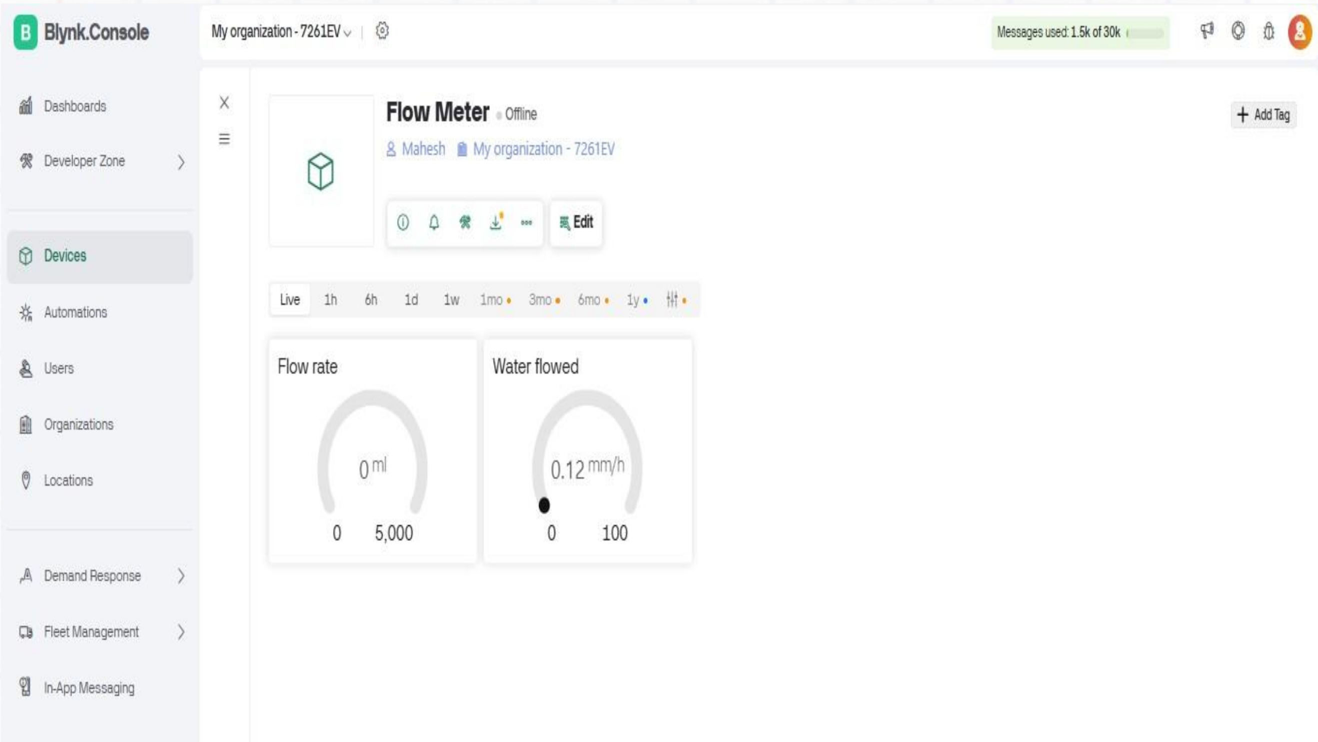
- Now Click Choose device >> Flow Meter and click Create



- Now everything is done, and you got BLYNK_AUTH_TOKEN Copy that Auth token on the top right part of the page and paste it into the BLYNK_AUTH_TOKEN variable, which is immediately below the Wi-Fi variables in the Arduino code

7. Upload the Code

- Copy the provided code into a new sketch in the Arduino IDE.
- Fill in your wifi connection's name and password within the empty quotation marks next to the YOUR_WIFI_SSID and YOUR_WIFI_PASS variables. Ensure the Wifi network has no firewalls, else the ESP32 won't connect to it.
- Click the 'Verify' button on the Arduino IDE - shown by a tick symbol. Once you receive the "Done compiling" message, click the 'Upload' button, shown as a right arrow near the 'Verify' button. Wait until the "Done uploading" message appears.
- After uploading the code, Click the magnifying glass icon on the top-right of the Arduino IDE. It will open the Serial Monitor. If the connection was successful.
- Now go to the Devices page of the blynk website. You will find a device named flowmeter containing variables flowrate and water flowed.



Code

```
#define BLYNK_TEMPLATE_ID            "TMPL3RZRBeDYU"
#define BLYNK_TEMPLATE_NAME          "IoT Flowmeter"
#define BLYNK_AUTH_TOKEN             "oEPjIiM-Rwuz-VSRTNlaHddsPbkjNHXm"

#include <WiFi.h>
#include <BlynkSimpleEsp32.h>
#include <EEPROM.h>

// Wi-Fi credentials
const char ssid[] = "Mahesh";
const char pass[] = "123456789";

// Water Flow Sensor pin
#define FLOW_SENSOR_PIN 4   // Adjust according to your wiring

BlynkTimer timer;

volatile unsigned long pulseCount = 0;
float flowRate = 0.0;          // L/min
float totalLiters = 0.0;
unsigned long lastMillis = 0;

// ISR for water flow sensor
void IRAM_ATTR pulseCounter() {
  pulseCount++;
}

void setup() {
  Serial.begin(115200);
  delay(100);

  // Connect to Blynk
  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

  // Setup water flow sensor pin
  pinMode(FLOW_SENSOR_PIN, INPUT_PULLUP);
  attachInterrupt(digitalPinToInterrupt(FLOW_SENSOR_PIN), pulseCounter,
RISING);

  // EEPROM for total water usage storage
  EEPROM.begin(32);
  EEPROM.get(0, totalLiters);
  if (isnan(totalLiters)) totalLiters = 0.0;

  lastMillis = millis();

  // Send data every 5 seconds
  timer.setInterval(5000L, sendFlowDataToBlynk);
}
```

```
void loop() {
  Blynk.run();
  timer.run();
}

void sendFlowDataToBlynk() {
  unsigned long currentMillis = millis();
  float intervalSec = (currentMillis - lastMillis) / 1000.0;
  lastMillis = currentMillis;

  // Calculate flow rate (assuming 450 pulses per liter, adjust for your
sensor)
  flowRate = (pulseCount / 450.0) / (intervalSec / 60.0);  // L/min
  totalLiters += (pulseCount / 450.0);

  // Serial output
  Serial.printf("Flow Rate: %.2f L/min | Total Liters: %.2f\n", flowRate,
totalLiters);

  // Send to Blynk
  Blynk.virtualWrite(V0, flowRate);
  Blynk.virtualWrite(V1, totalLiters);

  // Reset pulse count
  pulseCount = 0;

  // Save total liters to EEPROM
  EEPROM.put(0, totalLiters);
  EEPROM.commit();
}
```

Now verify the code on Arduino

Then upload the code into the ESP32.

Tasks

The Flow Rate value in the Serial Monitor should read zero. If the Water Flowed value isn't initially zero, press the reset button on the ESP32. Now slowly blow into the Flow Meter pipe and check whether the flowrate and water flowed variables change in the Serial Monitor. Go to the blynk Dashboard page - the meter gauges should reflect the change in values.

Real-world Application

You have now learnt how to build an IoT based Flow Meter and are ready to apply it in real-world scenarios! This Flow Meter can be used in several applications. Identify locations in your college or home where it can be installed - this could be a hand wash area, a distribution pipe of a water tank, or even a drinking water dispenser.
One very simple and practical use-case is in an RO water purifier. Have you ever wondered how an RO water purifier works? Using Reverse Osmosis technology, it concentrates impurities and sediments in one portion of the water - which is wasted - leaving the purified portion for drinking. But do you have any idea how much water is wasted? Taking permission from your college authorities, locate an RO water purifier in your college and install the Flow Meter in the pipe that drains out the wastewater. It must be installed in such a way that the electronics are protected from water, heat or physical damage. Make sure that there is access to Wi-Fi in that area. Track the wastage quantities over different periods of time - a day, a week, a month… - and note these values. Additionally, you can build one more Flow Meter or use the same to measure the amount of purified water consumed form the same RO system.
You will be surprised to know the amount of water wasted to purify just 1L of water!
Make a video presenting the installation of the Flow Meter and the results/insights you got about the RO system. Share it with the Build Club Community on the Discord Server!
Also share your own unique applications Real-world Application of the IoT Flow Meter!

Club Activity:

1)
2)